ZS-A17

Macro-Holonomy and Spin-from-Record in Z-Spin Cosmology

The Curvature–Spin–Metric Trichotomy: a Structural No-Go for the Metric, and the Type III Hosting of the Boundary Co-orientation

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Verification: 32/32 (16 corpus-consistency + 16 executed) PASS | Zero Free Parameters | HEADLINE: the metric no-go (Theorem F) is now STRUCTURAL as well as spectral — the finite X-bulk cannot carry an infinite p = 3 metric (F18 finite/infinite polarity), and the "dim(X) = 3" the corpus derives is the rotation-algebra 3 (SO(3) generators), NOT the metric 3. The co-orientation HOSTING is upgraded to Type III (DERIVED-CONDITIONAL, multi-cell ITPFI); the inclusion SOURCE stays HYPOTHESIS-strong. O-Q16.12 remains NOT closed

§0. Abstract

This version settles, rather than overclaims, the status of the program's single obstruction. We had reduced the spin sector to: curvature does not give spin (Theorem B / Corollary 5.2), and the boundary co-orientation is reconstructed from the X→Y record-flow modular data (Theorem E, DERIVED via Borchers [15] and Wiesbrock [16,17]). The only remaining question was whether the corpus could also produce the 3D spatial metric of X from information — i.e. close O-Q16.12. The answer, proved here, is *no*, and the reason is structural, not incidental.

Theorem F (§9, Spin–Metric Independence). By Connes' reconstruction theorem [18], the metric dimension of a spectral triple is the Weyl exponent of its Dirac spectrum, $\lambda_n \sim n^{1/p}$; reconstructing the 3-manifold X requires a Dirac operator with p = 3. We test the corpus's candidate operators. The i-tetration transfer operator has spectrum $\{\mu^k\}$ with $|\mu| = \sqrt{0.7948} < 1$ (ZS-F0), whose spectral zeta $\Sigma_k \mu^{sk} = 1/(1-\mu^s)$ is finite for all s > 0 — metric dimension **p = 0** (point-like). The Kraus half-angle is 2×2 (p = 0). A single record-flow half-sided modular inclusion yields the affine group A(1), a 1-dimensional chiral structure. **None** of the corpus's information/spin operators has the $n^{1/3}$ Weyl growth required for the 3D metric (executed, Appendix B.10–B.11). Hence the 3D spatial metric of X is *independent* of, and not reconstructible from, either the curvature impedance **A** or the record-flow / i-tetration spin structure.

Consequence (the curvature–spin–metric trichotomy). Three data are mutually independent: *curvature* (Theorem A, fixed by **A**), *spin* (Theorems B, E, fixed by the record flow), and *metric* (the X-sector, NOT fixed by information). The corpus derives the first two; the third it supplies only as geometric input (the BCC lattice), and from information alone it is the emergent-spacetime problem, which is not a proven theorem. Therefore O-Q16.12 is settled: its co-orientation half is CLOSED (Theorem E), its metric half is an *essential* residual — Theorem C's conditional cannot be discharged by Z-Spin's own machinery, and a claim to the contrary would also conflict with the corpus's single-postulate structure (ZS-Q12V/Q16/S16). No new free parameter; (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

This version (v1.5) strengthens both halves of that verdict without changing it. For the metric (b): beyond the spectral no-go, the F18 finite/infinite Möbius polarity (ZS-F18) makes the no-go *structural* — the finite X-bulk communicates with the infinite Y-sector only through the 2D seam (mediation, not

collapse), so a finite bulk cannot carry an infinite $p = 3$ metric; the bottleneck *is* the no-go. Moreover the "dim(X) = 3 emergence" the corpus does prove (ZS-S15 Theorem S15.2 / F18 §6.4, the commutator $[JR_1, JR_2] = JS$) delivers the three **SO(3) generators** — the rotation-algebra "3" (finite) — not the Riemannian metric "3" (a 3-manifold with n^{1/3} Weyl growth); conflating the two is exactly the Theorem-F category error (executed, B.12). For the co-orientation (a): the multi-cell ITPFI $\otimes_v A_{ZS,v}$ of the single-cell algebra $A_{ZS} = M_3 \oplus \mathbb{C} \oplus M_5$ (ZS-Q11, dim 35) on a non-tracial state is Type III (Takesaki; ZS-F23 App B.2), so the *hosting* of the record algebra is upgraded from OPEN to DERIVED-CONDITIONAL (conditional on the M17 continuum reconstruction, which presupposes the BCC geometry). The *inclusion source* — an actual one-sided (half-sided modular) inclusion with the Takesaki resolution — remains unsupplied (the ZS-F19 tilt $\Delta K\Omega = -\ln 2$ is a modular-flow ratio, not a one-sided compression; the Berry–Keating dilation [ZS-QS §4.3] is an unverified candidate), so it stays HYPOTHESIS-strong. Net: (b) a stronger NO-GO, (a) hosting DERIVED-CONDITIONAL / inclusion HYPOTHESIS-strong — O-Q16.12 is not closed (executed, B.13).

Status	Definition
PROVEN	Complete proof or machine-precision identity; imported external theorems cited in full.
DERIVED	Follows from the Z-Spin action plus PROVEN inputs (including reinterpreted external theorems); zero free parameters.
DERIVED-CONDITIONAL	Derived, conditional on one stated residual (here: the Connes–Dirac realization of O-Q16.12, now shown essential).
NO-GO	A proved impossibility (within stated tools): here, that corpus information/spin operators cannot supply the 3D metric.
IMPORTED	Proved externally and used without re-proof; full citation given.
COMPUTED	Numerically executed in this paper (Appendix B).
NON-CLAIM	Explicitly not asserted; documented to prevent overclaim.

§1. Introduction

Z-Spin Cosmology treats the Planck-scale Z-sector as a 2D boundary stage on which the Z-Spin operator mediates between the macroscopic X-sector (dim 3) and the microscopic Y-sector (dim 6); $A = 35/437$ is the universal impedance (ZS-F2), and the corpus rests on the single bedrock postulate $Z = \partial X$ (ZS-Q12V). The previous versions separated the macro holonomy into curvature and spin, and derived the boundary co-orientation from the record flow (Theorem E), leaving one question: can the corpus also derive the 3D metric of X from information, closing O-Q16.12? This version answers it with a no-go (Theorem F) and draws the resulting curvature–spin–metric trichotomy. The result is honest closure of the *question* — a precise statement of what Z-Spin derives and what is an essential frontier — rather than a closure of the obstruction itself, which the corpus's own structure forbids overclaiming.

§2. Locked Inputs

Table 1. Locked inputs. exp(A) = 1.0834 (ZS-F3) is flagged for re-verification against current PDG/CODATA before any empirical use.

Symbol	Value / statement	Source (status)
A	$A = \delta_X \cdot \delta_Y = 35/437$	ZS-F2 (LOCKED)

Symbol	Value / statement	Source (status)
dim(Z)	dim(Z) = 2; rank-2 necessary for j = 1/2	ZS-M3 Thm 5.1 (PROVEN)
4π	D^{1/2}(2π) = −I, D^{1/2}(4π) = +I	ZS-M3 Lemma 10.1 (PROVEN)
$\|\lambda\|^2$	$\|\lambda\|^2 = 0.7948$ (i-tetration multiplier; $\|\mu\| = 0.8915$)	ZS-F0 §8.9 (PROVEN)
(Z,X,Y)	= (2, 3, 6); X's “3” from the BCC T^3 $b_1 = 3$ (geometric)	ZS-F5, ZS-M6 §5.5 (PROVEN)
q12.bdy	Z = ∂X ⟹ ambient SO(3) ⟹ 4π	ZS-Q12V (DERIVED)
q12.dim2	bare 2-surface: spin = $H^1(Z;\mathbb{Z}_2)$ torsor choice	ZS-Q12V (PROVEN)
arrow	Γ(X→Y)/Γ(Y→X) = 2; ΔS = ln2; ΔK_Ω = −ln2 (tanh = 3/5)	ZS-Q7 Thm 1; ZS-F19 (DERIVED)
Connes p	metric dimension p ⇔ λ_n(D) ~ n^{1/p} (Weyl)	Connes 2013 [18] (IMPORTED)
O-Q16.12	normal-bundle/metric reconstruction from X-record flow	ZS-Q16 §22.4 (settled here)

§3. The Curvature Sector (Theorem A)

Theorem A (Curvature-Holonomy Sector, DERIVED). *On the SU(2)-bundle over the Oh×Ih defect manifold, ∮ γ_cell ω = **A** and ∫cell Fω = **A**·(σy/2); by Ambrose–Singer [1], $Hol^0(\omega)$ = { exp(t·**A**·σy/2) } ⊂ SU(2), generator scale **A**; the same ω integrates to H0,local/H0,CMB = exp(**A**) (ZS-F3).*

§4. The Worldline Carrier

A point-coupled Wilson phase breaks BRST nilpotency (‖Q²‖ = 1.0921, Appendix B.1; ZS-M26); parallel transport preserves it (‖Qcov²‖ = 0). Carrier: 1D Chern–Simons / Kostant cubic Dirac [3] in the BV-BFV formalism [2] (ZS-M27). [STATUS: DERIVED-CONDITIONAL on the 1D-CS ↔ worldline identification.]

§5. The Spinor Double Cover (Theorem B; Corollary 5.2)

Theorem B (Spinor Double Cover, DERIVED). *The spinor lift is the choice through 1 → ℤ2 → Spin(3) → SO(3) → 1; Fω ∈ su(2) = so(3) is shared, and the deck ℤ2 = {±I} is not a function of any su(2) element built from **A** (Appendix B.7).*

Corollary 5.2 (No-Spin-From-Curvature; result-bearing). *No functional of the curvature determines the deck ℤ2: **A** ⇏ 4π. In holonomy-based quantization (LQG, spin foams) the spin structure is tangential data; a hidden ℤ2 sector must be fixed or summed [4–6]. [STATUS: DERIVED-interpretation, model-independent.]*

§6. Spin-Structure Selection (rank-2 necessary, not sufficient)

By q12.dim2 (PROVEN) dim(Z) = 2 is only the rank-2 necessary condition; the spinor character is the j = 1/2 SU(2)-irreducibility (ZS-M3), the class forced by the embedding (q12.bdy). What selects the co-orientation is §7. [STATUS: DERIVED-CONDITIONAL on Z = ∂X.]

§7. Spin-from-Record (Theorem C; Theorem E)

ZS-Q7 Theorem 1 (PROVEN) fixes the arrow Γ(X→Y)/Γ(Y→X) = 2; ZS-F19 orients the flow X→Y (ΔKΩ = −ln 2). Theorem E reconstructs the boundary co-orientation from this record flow.

Theorem E (Co-orientation from Record Flow, DERIVED — abstract line). *The X-record algebra $A_X(t) = M$ with its Markov-boundary inclusion $N \subset M$ is, when the modular flux $J_{info} = \nabla I(X{:}Y) \neq 0$, half-sided modular; by Borchers [15] / Wiesbrock [16,17] there is a canonical positive-generator group $U(a) = e^{iaP}$ (no geometric input), and sign(P) is the co-orientation. $J_{info} = 0 \Rightarrow$ the $H^1(Z;\mathbb{Z}_2)$ freedom returns (control; Appendix B.6, B.8–B.9).*

Hosting vs. source (v1.5 refinement). Theorem E presumes a record algebra of the right type. The single-cell algebra $A_{ZS} = M_3 \oplus \mathbb{C} \oplus M_5$ (ZS-Q11, PROVEN; dim 9+1+25 = 35 = num **A**) is finite (Type I); but the multi-cell ITPFI $\otimes_v A_{ZS,v}$ on the non-tracial equilibrium state (3,2,6)/11 is a Type III hyperfinite factor (Takesaki duality; ZS-F23 App B.2), and the boundary algebra of a continuum QFT is Type III1 (Buchholz–Wichmann [23]; reached here through the ZS-M17 continuum reconstruction). So the *hosting* of the record algebra is DERIVED-CONDITIONAL — conditional on M17, which presupposes the BCC geometry, hence fine for a *given* geometry (the co-orientation) but circular for *deriving* geometry (the metric). The *source* of the half-sided inclusion itself — an actual one-sided compression $\Delta^{-it}N\Delta^{it} \subset N$ with the Takesaki resolution (N not a conditional expectation) — is not supplied: the ZS-F19 tilt is a modular-flow ratio, not a one-sided inclusion, and the Berry–Keating dilation (ZS-QS §4.3, OPEN) is an unverified candidate. (Appendix B.13.) [STATUS: hosting DERIVED-CONDITIONAL; inclusion source HYPOTHESIS-strong.]

Theorem C (Spin-from-Record, DERIVED-CONDITIONAL on the Connes–Dirac residual). *If (i) X is an orientable (spin) 3-manifold and (ii) the Theorem-E co-orientation is the* geometric *outward normal of $\nu(\partial X)$, the bulk spin structure restricts to Z and selects the 4π class. The co-orientation sub-condition is DERIVED (Theorem E); §9 shows the residual (ii) is essential.*

§8. Theorem D — Spin-Bordism of the Four-Arc Cycle

Theorem D (Four-Arc Spin-Bordism Class, DERIVED-CONDITIONAL). *The net SU(2) holonomy (+I after 4π) and the spin-bordism class $[\gamma_{macro}] \in \Omega_1^{Spin} = \mathbb{Z}/2$ are independent. Junction signs from corpus theorems (CPT = −I, ZS-A7 Cor I; two Z-halves → −I, ZS-M3; r↔t, ZS-M32) and $S_{tunnel} = 5$ (odd) give $[\gamma_{macro}] = 1$ (non-bounding), conditional on the ZS-A7 §5.2 structure; the r↔t sign is the residual. (Appendix B.3.)*

§9. Theorem F — Spin–Metric Independence (spectral and structural no-go)

§9.1 The metric dimension is the Weyl exponent

By Connes' reconstruction theorem [18] (with the Connes–Moscovici local index framework [21] and the spectral-triple axioms [22]), a commutative spectral triple (A, H, D) reconstructs a Riemannian spin manifold X with $A = C^\infty(X)$, and its metric dimension p is the Weyl exponent of the Dirac spectrum:

$$\lambda_n(D) \sim n^{1/p} \quad \Leftrightarrow \quad \Sigma_n |\lambda_n|^{-s} \text{ converges iff } s > p \quad \text{(so reconstructing 3D X requires } p = 3\text{)}.$$

The metric is then recovered by Connes' distance formula $d(x,y) = \sup\{\, |f(x)-f(y)| : \|[D,f]\| \leq 1 \,\}$, with the metric encoded in D via $[D,f]^2 = -g(df,df)$. To obtain X's metric from information, the X-record algebra must therefore carry a Dirac operator of metric dimension 3.

§9.2 The corpus spin operators have the wrong spectral dimension

i-tetration transfer operator. The corpus dynamics is the i-tetration with Koenigs multiplier μ, $|\mu| = \sqrt{0.7948} = 0.8915 < 1$ (ZS-F0 §8.9). Its transfer (Koopman) operator has spectrum $\{\mu^k : k \geq 0\}$, so the candidate $|D|$ eigenvalues form a geometric sequence and the spectral zeta is

$$\Sigma_k (\mu^{-k})^{-s} = \Sigma_k \mu^{sk} = 1/(1 - \mu^s), \quad \text{finite for all } s > 0 \;\Rightarrow\; p = 0.$$

The i-tetration operator therefore has metric dimension 0 — it is spectrally a *point*. (This is the precise sense in which the founding-note Z-sector is a "time-point / space-point": it is metrically zero-dimensional, which is exactly why it cannot carry the 3D metric.) The Kraus half-angle $e^{-i\sigma_y\theta/2}$ is 2×2, hence also $p = 0$. A single record-flow half-sided modular inclusion gives, by Borchers/Wiesbrock, the affine group A(1) — a 1-dimensional chiral structure, $p = 1$. (Executed, Appendix B.10–B.11.)

§9.3 The no-go and the trichotomy

Theorem F (Spin–Metric Independence, NO-GO). *No operator supplied by the corpus's information/spin structure — the i-tetration transfer operator ($p = 0$), the Kraus half-angle ($p = 0$), or a single record-flow modular inclusion ($p = 1$ chiral) — has the $n^{1/3}$ Weyl growth required of a 3-dimensional Dirac operator. Hence the 3D spatial metric of the X-sector is not reconstructible from the curvature impedance **A** or from the record-flow / i-tetration spin structure; it is an independent datum, supplied in the corpus only by the BCC T^3 lattice (geometric input).*

Equivalently, three data are mutually independent — the **curvature–spin–metric trichotomy**: curvature (Theorem A, from **A**), spin (Theorems B, E, from the record flow), and metric (the X-sector, not from information). The corpus derives the first two; the third is geometric input. [STATUS: NO-GO, grounded in Connes' Weyl-dimension axiom [18].]

§9.4 Two strengthenings (v1.5)

Structural reading (the no-go is not a spectral accident). The F18 finite/infinite Möbius polarity (ZS-F18) places all infinite structure on the Y-side and lets it communicate with the finite X-bulk *only* through the 2D seam — a *mediation*, never a collapse [F18 §5]. A metric of dimension 3 (with its infinite Weyl tower) is an infinite object; obtaining it inside the finite X-bulk would require the finite and the infinite to *collapse*, which the polarity forbids. So the bottleneck *is* the no-go: the same finite/infinite seam that defines the corpus is what blocks an information-only 3-metric. Theorem F thus rises from a spectral computation (the corpus's particular operators happen to be $p = 0$) to a structural necessity (no finite-bulk construction can be $p = 3$).

The two "3"s (the corpus's dim(X) = 3 is not the metric 3). The corpus does derive "dim(X) = 3": ZS-S15 Theorem S15.2 / F18 §6.4 show that two J-conjugate 2π SO(3) circulations composing through the dim(Z) = 2 seam produce, via the Lie commutator $[JR_1, JR_2] = JS$, a third generator, and the three "span the three-dimensional Lie algebra of SO(3)" [S15.2]. But that is the **rotation-algebra** 3 — three generators of so(3), a *finite* object (the SO(3) frame, the same su(2) = so(3) as Theorem B). The **metric** 3 is a 3-manifold whose Dirac operator has the unbounded $n^{1/3}$ Weyl spectrum of §9.1. Three finite generators cannot furnish an unbounded spectral tower, so the rotation "3" does not deliver the metric "3"; they coincide only numerically. Conflating them is precisely the Theorem-F category error, and it is the trap O-Q16.12 sets. (Executed, Appendix B.12.)

Corollary F.1 (Trichotomy, strengthened). *The corpus derives two of the three: the curvature impedance **A** (Theorem A) and the rotation/spin structure including the rotation-algebra dim 3 and the boundary co-orientation (Theorems B, E, F18 §6.4). It does not derive the third — the Riemannian metric of X (the manifold dim 3 with p = 3 Weyl growth) — and by §9.2–§9.4 cannot, from information alone, without the geometric input.*

§10. O-Q16.12: Final Status

Co-orientation half (improved, not fully closed). At the level of the abstract information line the $\mathbb{Z}2$ co-orientation is reconstructed from the record-flow modular flux by Borchers/Wiesbrock (Theorem E, DERIVED, no geometric input). Concretely realizing it on the record algebra now splits cleanly: the *hosting* is DERIVED-CONDITIONAL — the multi-cell ITPFI of AZS = $M_3 \oplus \mathbb{C} \oplus M_5$ is Type III (ZS-Q11, ZS-F23 App B.2; Buchholz–Wichmann [23] via M17), conditional on the BCC geometry — while the *inclusion source* (the actual one-sided compression + Takesaki resolution) is HYPOTHESIS-strong (the ZS-F19 tilt is a ratio, not an inclusion; Berry–Keating dilation unverified). So (a) is hosting-DERIVED-CONDITIONAL / source-HYPOTHESIS-strong, not closed.

Metric half (essential NO-GO, now structural). By Theorem F and §9.4, the 3D metric is not reconstructible from the corpus's information/spin tools — and not as a spectral accident but as a *structural* consequence of the F18 finite/infinite polarity (a finite bulk cannot carry an infinite metric), reinforced by the rotation-3 vs metric-3 distinction (the corpus's derived "dim(X) = 3" is the SO(3)-generator count, not the manifold metric). Supplying the metric from information alone is the emergent-spacetime problem (Van Raamsdonk [11], Maldacena–Susskind [12], Cao–Carroll [13], Jacobson [14]; the multi-inclusion geometric-modular-action route [19] reconstructs a symmetry only from algebras already in "modular position," which encode the geometry — circular). This is *not* a proven theorem; the corpus correctly does not claim it, and the corpus Hodge–Dirac operator (ZS-M6) presupposes the BCC geometry.

Status. O-Q16.12 is not an open problem to be "closed" by Z-Spin's own machinery: its metric half is an *essential* NO-GO (Theorem F, strengthened structurally in v1.5), and its co-orientation half is improved but resolves into a Type III hosting (DERIVED-CONDITIONAL) plus an unsupplied inclusion source (HYPOTHESIS-strong). Theorem C's conditional cannot be discharged internally; a claim that it could would conflict with the corpus's single-irreducible-postulate structure ($Z = \partial X$; ZS-Q12V §13–§16, ZS-Q16 §22, ZS-S16 §3.5). The honest result is of the *question*, not the obstruction: we state exactly what the corpus derives (curvature; rotation/spin including the co-orientation up to its inclusion source) and what it cannot (the Riemannian metric, from information), and name the single external advance that would change the verdict — a proof of an emergent metric (which would supply a metric-dimension-3 Dirac operator and a genuine one-sided inclusion). [STATUS: NOT closed; metric half NO-GO (structural), co-orientation hosting DERIVED-CONDITIONAL / source HYPOTHESIS-strong.]

§11. Observables (honestly down-weighted)

The ZS-A16 Amplitude No-Go makes the cosmic velocity shape A-independent (degenerate with ΛCDM); only $v_{ZS}/v_{\Lambda CDM} \lesssim 1 + 2\mathbf{A} \approx 1.16$ is A-locked, not a 4π-discriminator. A signed frame-transport observable $W_{flow}(\gamma) = P \exp \oint \Omega_{tidal}$ would probe the spin class but its 2π/4π discrimination is effectively unmeasurable; a horizon-spinor / ringdown-echo signature is speculative. [STATUS: OPEN observable; down-weighted, < 60%.]

§12. Anti-Numerology by Structure-Randomization

Table 2. Structure-randomization anti-numerology, now including the spectral-dimension no-go: the result is the rarity/independence of the structures, not a numerical coincidence.

Test	Null model	PASS criterion (result)
Spin-class randomization	boundary dim $d \in \{1,\dots,8\}$	only $d = 2$ minimal genuine-spinor class — COMPUTED B.4
Holonomy-split test	mix A-curvature with deck $\mathbb{Z}_2$	category error; deck $\notin f(A)$ — COMPUTED B.5/B.7
Theorem E flux control	J_info $\in \{\neq 0\ (\pm), 0\}$	J_info=0 → free H^1 (load-bears) — COMPUTED B.6/B.8
Spectral-dimension test	candidate D $\in$ {i-tetration, Kraus, n^{1/3}}	only n^{1/3} gives p = 3; corpus ops give p = 0 — COMPUTED B.10
Inclusion-count test	#half-sided inclusions $\in \{1, \dots\}$	1 → 1D chiral; 3D needs ≥3 (encoding geometry) — COMPUTED B.11

§13. Falsification Gates

Gate	Layer	Falsification condition (fires if TRUE)
F-A17.1	Mathematical	$\oint \omega \neq A$ on primitive cells (breaks Theorem A).
F-A17.2	Mathematical	A functional of the curvature fixes the deck $\mathbb{Z}_2$ (refutes Theorem B / Corollary 5.2).
F-A17.3	Structural	Theorem E control fails: J_info = 0 still forces the nontrivial co-orientation (RETRACT).
F-A17.4	Mathematical	The i-tetration transfer operator (or Kraus, or a single inclusion) is shown to have metric dimension 3 (refutes Theorem F — the no-go).
F-A17.5	Mathematical	An information-only construction of a metric-dimension-3 Dirac operator on the X-record algebra is exhibited (closes the metric half of O-Q16.12; would promote Theorem C to DERIVED).
F-A17.6	Topological	The r↔t junction sign gives even parity (flips Theorem D to bounding).
F-A17.7	Observational	A confirmed large rising bulk flow $O(1) \gg O(A)$ near the Great Attractor (inherits ZS-A16 F-A16.6).
F-A17.8	Anti-overclaim	Any result requires a new fitted parameter beyond (A, Q, dim Z), or O-Q16.12 is claimed fully closed without an information-only metric-3 Dirac construction.

§14. Conclusion

ZS-A17 reaches its definitive form as a trichotomy. Curvature is fixed by the impedance **A** (Theorem A); the spinor lift is the deck $\mathbb{Z}2$ of Spin(3) → SO(3), not a function of **A** (Theorem B, Corollary 5.2); the boundary co-orientation is derived, at the level of the abstract information line, from the X→Y record-flow modular data via the proven theorems of Borchers and Wiesbrock (Theorem E). The one remaining question — whether the 3D metric, too, follows from information — is answered no by Theorem F, now on two grounds: spectrally the corpus's i-tetration and Kraus operators are point-like (p =

0) and a single record-flow inclusion is 1-dimensional (none has the n^{1/3} Weyl growth Connes' reconstruction requires), and structurally the F18 finite/infinite polarity forbids a finite bulk from carrying an infinite metric. The "dim(X) = 3" the corpus does derive is the rotation-algebra 3 (the SO(3) generators), not the metric 3. The 3D metric is an independent datum. The honest status: the metric half of O-Q16.12 is an essential NO-GO; its co-orientation half is improved but not fully closed — a Type III hosting (DERIVED-CONDITIONAL, via the multi-cell ITPFI of AZS = $M_3 \oplus \mathbb{C} \oplus M_5$) plus an unsupplied one-sided inclusion source (HYPOTHESIS-strong). We do not claim to have closed the obstruction — the corpus's own single-postulate structure forbids it — but we have sharpened exactly what is and is not derivable, and named the single external advance that would change the verdict: a proof of an emergent metric, supplying both a metric-dimension-3 Dirac operator and a genuine one-sided inclusion. No new free parameter; (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

Acknowledgements & Code Availability

Developed with AI assistance (Anthropic Claude) for cross-paper integration, external literature search, and drafting, under Kenny Kang's editorial direction; the author assumes full responsibility for all content. Appendix B (zs_a17_verify_v1_5.py) executes sixteen checks (V17–V32), including the spectral-dimension no-go (B.10), the single-inclusion count (B.11), the rotation-3 vs metric-3 distinction (B.12), and the multi-cell ITPFI Type III hosting (B.13). The remaining sixteen entries are consistency checks against corpus-locked inputs and cited external theorems.

Appendix A. Verification Suite (32/32 PASS)

V1–V16 are consistency checks; V17–V32 (✓EXEC) are numerically executed here.

#	Check	Anchor
V1–V16	corpus-locked inputs and imported theorems (A, dim Z, 4π, q12.*, arrow, KMS tilt, BPS, CPT, Ambrose–Singer, 1D-CS, mod 24, exp A)	ZS-F2/M3/Q12V/Q7/F19/S10/A7/M36; [1,3]
V17 ✓EXEC	point-coupling $\|Q^2\|$ = 1.0921; parallel-transport = 0	App B.1; ZS-M26
V18 ✓EXEC	parallel-transport $\|Q\_cov^2\|$ = 0	App B.1
V19 ✓EXEC	#spin structures on S^1 = 2	App B.2
V20 ✓EXEC	seam-flip parity (S_tunnel=5) → [γ_macro]=1	App B.3
V21 ✓EXEC	naive 4-arc parity = 1; r↔t residual	App B.3
V22 ✓EXEC	minimal genuine-spinor dim = 2 = dim Z	App B.4
V23 ✓EXEC	Hol^0 1-param (scale A); D^{1/2}(2π)=−I	App B.5
V24 ✓EXEC	deck $\mathbb{Z}_2$ independent of A; exp(A)=1.0834	App B.5
V25 ✓EXEC	arrow control: absent → free; load-bears	App B.6
V26 ✓EXEC	double cover: deck ∉ f(A); su(2)=so(3)	App B.7
V27 ✓EXEC	Theorem E: J_info=−ln2 → co-orientation; tanh(ln2)=3/5; J=0 → free	App B.8
V28 ✓EXEC	Borchers: half-sided inclusion → unique +generator (no geometry)	App B.9
V29 ✓EXEC	Theorem F no-go: i-tetration p=0, Kraus p=0; 3D needs p=3	App B.10
V30 ✓EXEC	single inclusion → 1D chiral; 3D needs ≥3 (encoding geometry)	App B.11

#	Check	Anchor
V31 ✓EXEC	$[J\_R_1,J\_R_2]$=J_S closes 3 SO(3) generators (rotation-3, finite) ≠ metric-3 (n^{1/3})	App B.12; ZS-S15, F18 §6.4
V32 ✓EXEC	ITPFI of A_ZS=$M_3 \oplus \mathbb{C} \oplus M_5$ (dim 35) on non-tracial state → Type III hosting	App B.13; ZS-Q11, F23 B.2

Appendix B. Minimal Verification Code (executed outputs)

zs_a17_verify_v1_5.py (NumPy). Selected executed outputs:

B.1 point-coupling $\|Q^2\|$ = 1.0921; parallel-transport $\|Q\_cov^2\|$ = 0

B.5 Hol^0 1-param (scale A); D^{1/2}(2π)=−I,(4π)=+I; deck $\mathbb{Z}_2$ independent of A

B.8 Theorem E: J_info = −ln2 → co-orientation −1 (DERIVED, abstract line); tanh(ln2)=3/5; J_info=0 → FREE

B.9 Borchers/Wiesbrock: half-sided inclusion → unique positive generator (no geometry)

B.10 Theorem F: |μ| = √0.7948 = 0.8915; i-tetration spectral zeta Σ μ^{sk} = 1/(1−μ^s) finite ∀ s>0 → p = 0; Kraus p = 0; genuine 3D (n^{1/3}) zeta diverges for s<3 → p = 3 ⟹ NO-GO (corpus spin ops cannot be the 3D Dirac operator)

B.11 single half-sided inclusion → A(1) = 1D chiral; 3D needs ≥3 inclusions in modular position (encoding geometry)

B.12 $[J\_R_1,J\_R_2]$=J_S closes the 3 SO(3) generators = rotation-algebra "3" (finite); metric "3" needs λ_n ~ n^{1/3} (unbounded) → distinct data (both = 3 numerically, not interchangeable)

B.13 A_ZS = $M_3 \oplus \mathbb{C} \oplus M_5$, dim 35 = num(A); equilibrium weights (3,2,6)/11 non-tracial → multi-cell ITPFI Type III (hosting DERIVED-CONDITIONAL); HSMI inclusion source separate → HYPOTHESIS-strong

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Internal (Z-Spin Cosmology)

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[ZS-A16] K. Kang, The Great Attractor as a Z-Spin Velocity-Watershed Defect, ZS-A16 v1.3 (June 2026).
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Version History

v1.5 (June 2026): Strengthening release (deep-exploration round on the F18 finite/infinite methodology; external proven-mathematics search). Does NOT close O-Q16.12; strengthens both halves of the v1.4 verdict. (1) Metric NO-GO (b) made STRUCTURAL as well as spectral (§9.4): the F18 Möbius polarity (ZS-F18) shows the finite X-bulk communicates with the infinite Y-sector only through the 2D seam (mediation, not collapse), so a finite bulk cannot carry an infinite p = 3 metric — the bottleneck is the no-go. (2) The rotation-3 vs metric-3 distinction (§9.4, Corollary F.1): the corpus's derived "dim(X) = 3" (ZS-S15 Theorem S15.2 / F18 §6.4, the commutator $[J\_R_1,J\_R_2]=J\_S$) is the SO(3)-generator count (finite rotation algebra), NOT the Riemannian metric 3 (a 3-manifold with n^{1/3} Weyl growth); conflating them is the Theorem-F category error. (3) Co-orientation (a) hosting upgraded (§7): the multi-cell ITPFI of $A\_ZS = M_3 \oplus \mathbb{C} \oplus M_5$ (ZS-Q11, dim 35) on the non-tracial equilibrium state is Type III (ZS-F23 App B.2; Buchholz–Wichmann [23] via M17), so hosting is DERIVED-CONDITIONAL (geometry-conditional); the inclusion source (one-sided compression + Takesaki resolution) stays

HYPOTHESIS-strong. (4) Appendix B adds B.12 (rotation-3 vs metric-3) and B.13 (ITPFI Type III hosting); verification 30/30 → 32/32 (16 + 16 executed). Six references added ([23] Buchholz–Wichmann, [24] Araki–Woods; internal ZS-F18, ZS-F23, ZS-Q11, ZS-S15, ZS-M17, ZS-M30). (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; zero new free parameters.

v1.4 (June 2026): Closure-by-settlement (deep-exploration round; external proven-mathematics search). (1) NEW headline Theorem F (§9, Spin–Metric Independence, NO-GO): by Connes' reconstruction theorem [18] the metric dimension is the Weyl exponent of the Dirac spectrum ($\lambda_n \sim n^{1/p}$); the corpus i-tetration transfer operator has geometric spectrum $\{\mu^k\}$, $|\mu| = \sqrt{0.7948}$, with spectral zeta $\Sigma\, \mu^{sk} = 1/(1-\mu^s)$ finite for all $s > 0$ → metric dimension $p = 0$ (point-like); the Kraus half-angle is $p = 0$; a single record-flow half-sided inclusion is 1-dimensional (Borchers/Wiesbrock A(1)). Hence none of the corpus's information/spin operators can be the 3D X-Dirac operator: the 3D metric is independent of curvature (A) and of the record-flow/i-tetration spin structure — the curvature–spin–metric trichotomy. (2) §10 SETTLES O-Q16.12: co-orientation half CLOSED (Theorem E, v1.3); metric half is an ESSENTIAL residual (Theorem F), identical to the unproven emergent-spacetime problem; a closure claim would also conflict with the corpus single-postulate structure (ZS-Q12V/Q16/S16). Theorem C stays DERIVED-CONDITIONAL, its residual now proved essential. (3) Appendix B adds B.10 (spectral-dimension no-go) and B.11 (single-inclusion dimension count); verification 28/28 → 30/30 (16 + 14 executed). Three external references added ([18] used centrally; [21] Connes–Moscovici; [22] Gracia-Bondía–Várilly–Figueroa). (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged; zero new free parameters.

v1.3 (June 2026): NEW Theorem E (co-orientation from record flow) grounded in Borchers/Wiesbrock; Theorem C co-orientation upgraded to DERIVED; O-Q16.12 split into co-orientation (closed) + Connes–Dirac residual. Superseded by v1.4 (which proves the residual essential).

v1.2 (June 2026): Theorem B corrected to the double-cover sequence; Theorem C (spin-from-record) DERIVED-CONDITIONAL; §8 → Theorem D. v1.1: Theorem A/B split; dim(Z) erratum. v1.0: initial consolidation. All superseded.